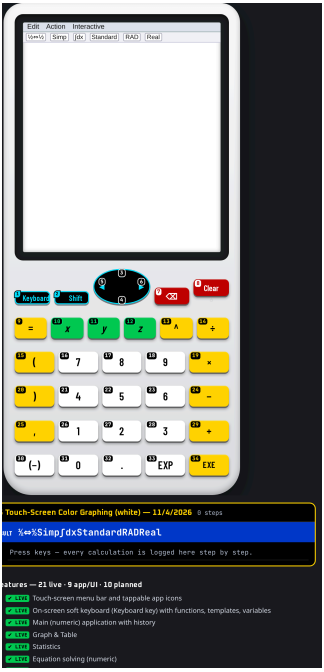
U-45 CALCULATORS

User Manual

Calculator #5 Touch-Screen Color Graphing (white)

Graphing – colour



Complete kit: functions · numbered key map · worked examples · tricks & hacks · shortcuts · common mistakes · practice drills

About this manual

This manual belongs to the **U-45 CALCULATORS** app and describes calculator **#5 — Touch-Screen Color Graphing (white)** exactly as it behaves in the app. Every worked example and every drill answer in this book was run on the app's own keys by an automated checker before printing; only results that matched were kept.

How key presses are written

Keys are written as a chip showing the **key number** and its face, for example [#1 Keyboard]. Numbers follow the numbering printed on the keys (switch it on with "# On"). A shifted function is written as two chips: the shift key, then the key that carries the function.

Entry style of this model

algebraic with history (type the expression, then EXE / ENTER).

Chapters

1. Getting started
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7. Keyboard shortcuts
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1 · Getting started

Open U-45 CALCULATORS, choose **#5** from the home screen (or the jump list at the top), and the calculator appears with its numbered, colour-coded keys. Click or tap keys, or type on a keyboard.

Numbers on every key

Every key carries a small number in the same reading order as this manual (left → right, top → bottom; the navigation pad counts ▲ ▼ ◀ ▶ then its centre). Use "# On / # Off" in the bar above the calculator to show or hide them — the setting is remembered.

High-contrast colours

Keys are coloured by job: white = digits, yellow = operators/execute, blue = functions, green = memory/variables, red = clear/delete/power, black with cyan = shift/modes/menus/navigation. Press "Classic" to return to the original model colours and "Hi-Contrast" to come back.

Three typefaces

Display text uses the U45 Display face, key legends use U45 Keys (a condensed face that stays readable on small keys) and key numbers / helper text use U45 Mono. All three are bundled inside the app, so they work offline.

Copy the display

The Copy button copies exactly what the display shows (indicators that are switched off are skipped) so you can paste a result into a message or spreadsheet.

Fit or 1:1

Fit scales the calculator to your screen; 1:1 shows it at its design size. On a phone, Fit is the comfortable choice.

Reset

Reset rebuilds this calculator from scratch: display, memory, stack and modes return to their start-up state. Use it before each worked example in this manual.

Keyboard

On a computer you can type: digits, + − * / ^ ( ), Enter or = to execute, Backspace to delete, Esc to clear, arrows to move. Letters reach common functions (see the Keyboard shortcuts chapter).

Tooltips

Hover over (or long-press) a key to see "#number name".

Tools rail (v1.4)

A cyan icon rail on the right of every calculator opens 15 panels: Currency (166 currencies, rates auto-updated from open.er-api.com or Frankfurter when online and cached with a date; the bundled dated snapshot is used offline), Units (exact SI factors), Health (published formulas, not medical advice), Simple, Complex, Loans, Construction, Mechanical and Engineering (30+ numbered tools each), Calendar generator, Customize (26 options, saved per calculator), Settings (31 options), History, Hide and View. On narrow phones the rail shows icons only.

History (v1.4)

Every calculation is logged per calculator with a date and time. Open History on the rail to search, re-use (types the input back into the calculator), copy, delete one, select several and delete, or delete all (asks for confirmation). History is stored encrypted on this device (AES-GCM); exporting it needs the passcode.

Camera Solve (v1.4)

The green CS button between Restore and Social reads a problem from the camera, a photo, a PDF or a text file. Recognition (Tesseract OCR, English) runs offline on the device; an online language download is only used if you switch it on in Settings. Check and edit the recognised text, then Solve shows the steps offline. OCR is not available when the app is opened as a plain file:// page in a browser.

Layouts and view modes (v1.4)

The layout selector between Fit and # offers Auto plus 12 layouts: 3D classic, 3D high-contrast, Classic, Pocket, Compact, Expanded, Desktop wide, Split, One-hand left, One-hand right, Large print and Minimal. Auto picks one for your screen and your choice is remembered per calculator. The View panel hides the top bars, the tool bar, the rail or everything (full screen); the round yellow ⤢ button at bottom-left (or Esc) always restores the normal view.

Hide items (v1.4)

The Hide panel lists every button, rail icon and section of this calculator as tick boxes. Hiding is instant; unhiding asks for the app passcode.

Image protection (v1.4)

Right-click, drag and long-press saving of images is blocked inside the app. Saving or sharing an image export (PNG, JPG, QR, calendar), printing a calendar, and downloading or sharing the app itself ask for the passcode, and exported images carry a U-45 watermark. Honest note: this cannot stop a screenshot or screen recording made by the operating system or another device.

Works offline

U-45 CALCULATORS is an offline app: after the first load everything — including fonts — comes from the device.

Colour groups on this calculator

Colour	Group	Keys on this model
num	Digits & number entry	#16, #17, #18, #21, #22, #23, #26, #27, #28, #30, #31, #32, #33
op	Operators & execute	#9, #13, #14, #15, #19, #20, #24, #25, #29, #34
fn	Functions	—
mem	Memory, variables & registers	#10, #11, #12
clr	Clear, delete & power	#7, #8
mod	Shift, modes, menus & navigation	#1, #2, #3, #4, #5, #6

2 · Features

40 features: 21 live on the keys, 9 app/UI features, 10 planned (documented but not yet working — they are clearly marked and never used in examples).

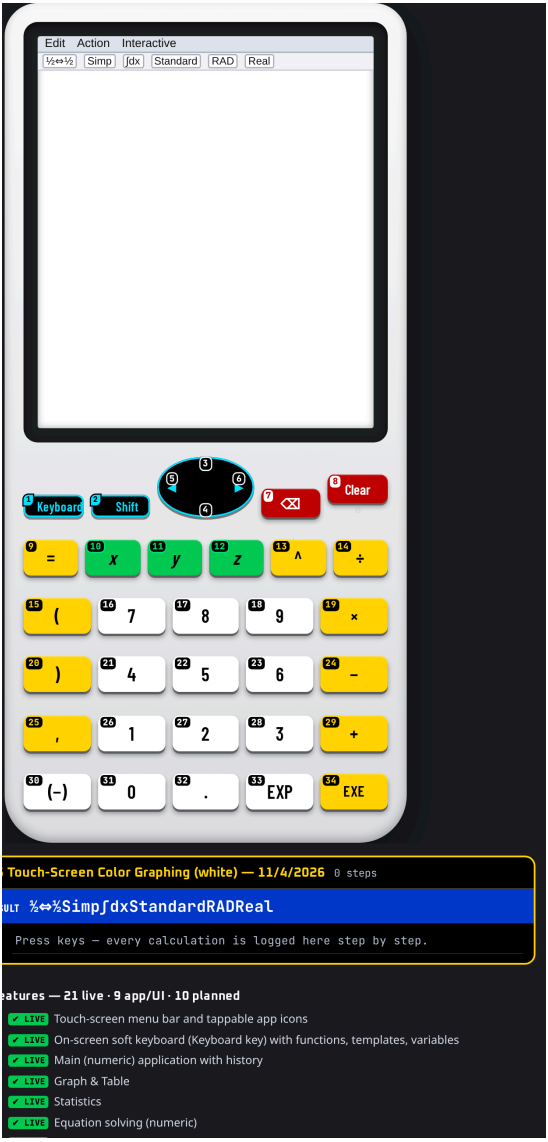
1. LIVE **Touch-screen menu bar and tappable app icons**
2. LIVE **On-screen soft keyboard (Keyboard key) with functions, templates, variables**
3. LIVE **Main (numeric) application with history**
4. LIVE **Graph & Table**
5. LIVE **Statistics**
6. LIVE **Equation solving (numeric)**
7. LIVE **Variables x, y, z on hard keys**
8. LIVE **Numeric Solver app**
9. LIVE **Matrix app**
10. LIVE **Distribution app**
11. LIVE **Power x^y / ^ with right-to-left chaining (2^3^2 = 512)**
12. LIVE **Scientific-notation entry $\times 10^x$ / EXP / EEX**
13. LIVE **Sign change (−) / +/− / CHS**
14. LIVE **SOLVE — numeric equation solver (Newton / bisection)**
15. LIVE **TABLE — tabulate f(x) over start / end / step**
16. LIVE **Statistics: data entry, n, $\bar{x}$, σ, s, Σx, Σx^2**
17. LIVE **Two-variable statistics and linear regression (a, b, r)**
18. LIVE **Matrix mode: define, add, multiply, det, transpose, identity**
19. LIVE **Backspace / DEL editing of the current entry**

20. LIVE Replay ▲ ▼ — scroll back to previous calculations and re-edit them
21. LIVE Probability distributions (normal, binomial, Poisson) and PRB menu
22. APP/UI Every key numbered 1…N in reading order — matches the key map in the user manual
23. APP/UI High-contrast 6-colour key groups (digits, operators, functions, memory, clear, modes) with a "Classic" toggle
24. APP/UI Three bundled offline fonts: Oxanium (display/titles), Barlow Condensed (key faces), JetBrains Mono (key numbers & helpers)
25. APP/UI Show / hide key numbers with one button (remembered between sessions)
26. APP/UI Copy the current display text to the clipboard
27. APP/UI Physical-keyboard shortcuts: digits, + – * / ^ ( ) . Enter Backspace Esc arrows, s c t l n r ! % e p q i
28. APP/UI Fit / 1:1 zoom that never pushes keys off a phone screen
29. APP/UI Reference photo viewer and tool-tip name on every key
30. APP/UI Reset = fresh power-on state; works fully offline (PWA cache / Android APK)
31. PLANNED Symbolic CAS algebra (solve, factor, expand)
32. PLANNED Geometry app
33. PLANNED eActivity
34. PLANNED Spreadsheet
35. PLANNED Stylus drawing
36. PLANNED Graph image export (PNG)
37. PLANNED Shareable function list link
38. PLANNED Dark screen theme

39. PLANNED **Speech read-out of values**

40. PLANNED **Exam mode**

3 · Numbered key map



Calculator #5 with key numbers shown (Hi-Contrast colours).

Key table

“Shift” = 2nd / f / SHIFT layer; “Alpha” = ALPHA layer; “3rd” = 3rd / g layer. A dot means the layer is empty on that key.

#	Face	Shift	Alpha	3rd	Group
1	Keyboard	·	·	·	mod
2	Shift	·	·	·	mod
3	▲	·	·	·	mod
4	▼	·	·	·	mod
5	◀	·	·	·	mod
6	▶	·	·	·	mod
7	⌫	·	·	·	clr
8	Clear	·	·	·	clr
9	=	·	·	·	op
10	x	·	·	·	mem
11	y	·	·	·	mem
12	z	·	·	·	mem
13	^	·	·	·	op
14	÷	·	·	·	op
15	(	·	·	·	op
16	7	·	·	·	num
17	8	·	·	·	num
18	9	·	·	·	num
19	×	·	·	·	op
20	)	·	·	·	op
21	4	·	·	·	num
22	5	·	·	·	num
23	6	·	·	·	num

#	Face	Shift	Alpha	3rd	Group
24	−	·	·	·	op
25	,	·	·	·	op
26	1	·	·	·	num
27	2	·	·	·	num
28	3	·	·	·	num
29	+	·	·	·	op
30	(−)	·	·	·	num
31	0	·	·	·	num
32	.	·	·	·	num
33	EXP	·	·	·	num
34	EXE	·	·	·	op

4 · Key-by-key reference

Every key on calculator #5, in number order, with every function it carries in this app.

#1 Keyboard `Shift, modes, menus & navigation`

Primary: On-screen keyboard panel

#2 Shift `Shift, modes, menus & navigation`

Primary: SHIFT / 2nd — the next key uses its upper (shift-coloured) function

#3 ▲ `Shift, modes, menus & navigation`

Primary: Cursor ▲ — up / previous history line / previous item

#4 ▼ `Shift, modes, menus & navigation`

Primary: Cursor ▼ — down / next history line / next item

#5 ◀ `Shift, modes, menus & navigation`

Primary: Cursor ◀ — move left in the line

#6 ▶ `Shift, modes, menus & navigation`

Primary: Cursor ▶ — move right in the line

#7 ⌫ Clear, delete & power

Primary: DEL — delete the character at the cursor

Keyboard: Backspace Delete

#8 Clear Clear, delete & power

Primary: ON / AC — power on, or clear all

#9 = Operators & execute

Primary: Equals — complete the calculation

#10 x Memory, variables & registers

Primary: Variable X (store with STO/ →, recall by name)

#11 y Memory, variables & registers

Primary: Variable Y (store with STO/ →, recall by name)

#12 z Memory, variables & registers

Primary: Variable Z (store with STO/ →, recall by name)

#13 ^ Operators & execute

Primary: Power x^y / ^ — raise to any power. Chains right-to-left (2^3^2 = 2^9 = 512)

Keyboard: ^

#14 ÷ Operators & execute

Primary: Division ÷

Keyboard: /

#15 (Operators & execute

Primary: Opening bracket (

Keyboard: (

#16 7 Digits & number entry

Primary: Digit 7

Keyboard: 7

#17 8 Digits & number entry

Primary: Digit 8

Keyboard: 8

#18 **9** `Digits & number entry`

Primary: Digit 9

Keyboard: 9

#19 **×** `Operators & execute`

Primary: Multiplication ×

Keyboard: * x

#20 **)** `Operators & execute`

Primary: Closing bracket). Closing brackets at the end of a line may be omitted

Keyboard:)

#21 **4** `Digits & number entry`

Primary: Digit 4

Keyboard: 4

#22 **5** `Digits & number entry`

Primary: Digit 5

Keyboard: 5

#23 **6** `Digits & number entry`

Primary: Digit 6

Keyboard: 6

#24 **–** `Operators & execute`

Primary: Subtraction –

Keyboard: -

#25 **,** `Operators & execute`

Primary: Comma / argument separator (between function arguments, list items or coordinates)

Keyboard: ,

#26 **1** `Digits & number entry`

Primary: Digit 1

Keyboard: 1

#27 **2** `Digits & number entry`

Primary: Digit 2

Keyboard: 2

#28 **3** `Digits & number entry`

Primary: Digit 3

Keyboard: 3

#29 + Operators & execute

Primary: Addition +

Keyboard: +

#30 (–) Digits & number entry

Primary: Negative sign (–) — makes the next number negative. Different from the subtraction key

#31 0 Digits & number entry

Primary: Digit 0

Keyboard: 0

#32 . Digits & number entry

Primary: Decimal point. Pressing it first gives "0."; a second point in the same number is ignored

Keyboard: .

#33 EXP Digits & number entry

Primary: ×10ˣ / EXP / EE — enter a power-of-ten exponent (scientific notation), e.g. 6.02 ×10ˣ 23

Keyboard: e

#34 EXE `Operators & execute`

Primary: Execute / = — evaluate the line and show the result (EXE / ENTER / =)

Keyboard: `Enter` `=`

5 · Worked examples

11 examples, each run on this calculator's keys by the U-45 checker. Follow the chips in order.

Example 5.1 — Order of operations

Calculate 12 + 3 × 4. Press Reset first.

#26 1 → #27 2 → #29 + → #28 3 → #19 × → #21 4 → #34 EXE

Display shows 24 **— expected 24. ✔ verified on the app.**

Multiplication is done before addition, so the calculator holds the + until the × part is finished.

Example 5.2 — Brackets first

Calculate (12 + 8) ÷ 4. Press Reset first.

#15 (→ #26 1 → #27 2 → #29 + → #17 8 → #20) → #14 ÷ → #21 4 → #34 EXE

Display shows 5 **— expected 5. ✔ verified on the app.**

Brackets force the sum to be worked out before the division.

Example 5.3 — Two brackets multiplied

Calculate (3 + 4) × (9 − 2). Press Reset first.

#15 (→ #28 3 → #29 + → #21 4 → #20) → #19 × → #15 (→ #18 9 → #24 − → #27 2 → #20) → #34 EXE

Display shows `49` **— expected 49. ✔ verified on the app.**

Both brackets are evaluated first, then multiplied.

Example 5.4 — Powers

Calculate 2^10. Press Reset first.

#27 2 → #13 ^ → #26 1 → #31 0 → #34 EXE

Display shows `1024` **— expected 1024. ✔ verified on the app.**

The power key raises the first number to the second.

Example 5.5 — Negative numbers

Calculate −5 + 12. Press Reset first.

#30 (−) → #22 5 → #29 + → #26 1 → #27 2 → #34 EXE

Display shows `7` **— expected 7. ✔ verified on the app.**

Use the sign key for a negative number — not the subtraction key.

Example 5.6 — Scientific notation

Calculate 6×10^2 × 3. Press Reset first.

#23 6 → #33 EXP → #27 2 → #19 × → #28 3 → #34 EXE

Display shows 1800 **— expected 1800. ✔ verified on the app.**

The exponent key enters ×10ⁿ without typing × 10 ^.

Example 5.7 — Continue from the answer

Calculate 100 × 4 = , then Ans ÷ 2. Press Reset first.

#26 1 → #31 0 → #31 0 → #19 × → #21 4 → #34 EXE → #14 ÷ → #27 2 → #34 EXE

Display shows 200 **— expected 200. ✔ verified on the app.**

Pressing an operator straight after a result continues from that answer (Ans).

Example 5.8 — Average of three numbers

Calculate (4 + 8 + 15) ÷ 3. Press Reset first.

#15 (→ #21 4 → #29 + → #17 8 → #29 + → #26 1 → #22 5 → #20) → #14 ÷ → #28 3 → #34 EXE

Display shows 9 **— expected 9. ✔ verified on the app.**

Add up the values, then divide by how many there are.

Example 5.9 — Mixed operations

Calculate 50 − 4 × 6 + 7. Press Reset first.

#22 5 → #31 0 → #24 − → #21 4 → #19 × → #23 6 → #29 + → #16 7 → #34 EXE

Display shows 33 **— expected 33. ✔ verified on the app.**

Only the × part is done first; the − and + are then worked left to right.

Example 5.10 — Dividing twice

Calculate 360 ÷ 4 ÷ 9. Press Reset first.

#28 3 → #23 6 → #31 0 → #14 ÷ → #21 4 → #14 ÷ → #18 9 → #34 EXE

Display shows 10 **— expected 10. ✔ verified on the app.**

Division is worked left to right: the first quotient is divided again.

Example 5.11 — Brackets inside a longer formula

Calculate ((3 + 5) × 4 − 2) ÷ 10. Press Reset first.

#15 (→ #15 (→ #28 3 → #29 + → #22 5 → #20) → #19 × → #21 4 → #24 − → #27 2 → #20) → #14 ÷ → #26 1 → #31 0 → #34 EXE

Display shows 3 — expected 3. ✔ verified on the app.

The innermost brackets are worked out first, then the outer brackets, then the division.

6 · Tricks & hacks

Hack 1: History reuse

Scroll up in the home screen history and press ENTER/EXE to copy an old entry or answer.

Hack 2: Ans chaining

Start an entry with an operator to continue from the previous answer.

Hack 3: Store variables

Use STO▶ / → to store values into letters A–Z, then use them in expressions.

Hack 4: Table of values

Use the TABLE view to scan where a function changes sign before solving.

Hack 5: Window first

If a graph looks empty, adjust the WINDOW / zoom standard.

Hack 6: Edit instead of retyping

After a result, press ◀ or ▶ to bring the last expression back with the cursor in it; change one number and press = again.

Hack 7: Ans chains

Start a line with an operator (e.g. ×) and the calculator inserts Ans automatically — great for compound growth: 1.05 ×, then = = = ...

Hack 8: Exact vs decimal

Natural-display models keep fractions, √ and π exact. Press S⇔D to flip to decimal; press it again to return.

Hack 9: Close brackets for free

Trailing closing brackets can be left off; = closes them for you.

Hack 10: Implicit multiplication

Type 2π or 3(4+1) without ×. Implicit multiplication binds tighter than ÷ on textbook models — use brackets if in doubt.

Hack 11: Store constants

Store a value you reuse (e.g. g = 9.81) with STO and a letter; then type the letter in formulas.

Hack 12: Power chains

x^y chains right to left: 2^3^2 = 2^9 = 512, not 64. Use brackets: (2^3)^2 = 64.

Hack 13: Angles

Check the D / R / G indicator before trig. sin 30 = 0.5 only in degrees.

Hack 14: Memory sums

M+ / M− keep a running total separate from the calculation line; RCL M shows it.

Hack 15: Replay history

Use ▲ ▼ to scroll earlier lines; press = on an old line to re-run it.

Hack 16: Find any key by number

Use Appendix B (function finder): look up the function, read its key number, then find that number on the keypad. Numbers run left → right, top → bottom from #1 to #34.

Hack 17: Type faster on a computer

22 keys on this model have a keyboard shortcut (see chapter 7). Typing digits and operators is much faster than clicking.

Hack 18: Check before you trust

Estimate first (round each number) and compare with the display. If the answer is 10× or 100× off, a decimal point or percent key was missed.

Hack 19: Copy results out

Press Copy in the bar above the calculator to put the display text on the clipboard instead of retyping it.

Hack 20: Colour blind or low vision?

Hi-Contrast colours use black/white/yellow/blue/green/red with black or white text for maximum contrast; key numbers stay on in both colour schemes.

7 · Keyboard shortcuts

With the calculator open on a computer, these physical keys press the numbered keys below (Ctrl/Alt/⌘ combinations are left to the browser). Shifted functions are reached by clicking the shift key first.

Keyboard	#	Key	Does
Backspace Delete	#7	⌫	DEL — delete the character at the cursor
^	#13	^	Power x^y / ^ — raise to any power. Chains right-to-left (2^3^2 = 2^9 = 512)
/	#14	÷	Division ÷
(	#15	(	Opening bracket (
7	#16	7	Digit 7
8	#17	8	Digit 8
9	#18	9	Digit 9
* x	#19	×	Multiplication ×
)	#20	)	Closing bracket). Closing brackets at the end of a line may be omitted
4	#21	4	Digit 4
5	#22	5	Digit 5
6	#23	6	Digit 6
-	#24	−	Subtraction −
,	#25	,	Comma / argument separator (between function arguments, list items or coordinates)
1	#26	1	Digit 1
2	#27	2	Digit 2
3	#28	3	Digit 3
+	#29	+	Addition +
0	#31	0	Digit 0

.	#32	.	Decimal point. Pressing it first gives “0.”; a second point in the same number is ignored
e	#33	EXP	×10ˣ / EXP / EE — enter a power-of-ten exponent (scientific notation), e.g. 6.02 ×10ˣ 23
Enter =	#34	EXE	Execute / = — evaluate the line and show the result (EXE / ENTER / =)

App shortcuts

“# On / # Off”	show / hide key numbers
“Classic / Hi-Contrast”	switch key colours
“Copy”	copy display text
“Fit / 1:1”	scale to screen / design size
“Reset”	restart this calculator
Jump list	switch to another of the 45 calculators

8 · Common mistakes

Mistake 1: (–) vs –

Use the negative key for negative numbers.

Mistake 2: Radian mode

Graphs of sin x in degrees look flat over −10..10; switch to radians for standard trig windows.

Mistake 3: Implicit multiplication

2/3x may be read as 2/(3x) depending on the model; use brackets.

Mistake 4: Using – instead of (–)

The negative key (–) and the subtraction key are different; "– 5" at the start uses Ans – 5.

Mistake 5: Wrong angle unit

If sin 30 gives −0.988, the calculator is in radians. Switch to degrees in SETUP.

Mistake 6: -3^2 vs $(-3)^2$

$-3^2 = -9$ because the square is applied first. Type $(-3)^2$ to get 9.

Mistake 7: Division precedence

1 ÷ 2π may be read as 1 ÷ (2π) on textbook models. Type 1 ÷ (2 × π) explicitly to be sure.

Mistake 8: Forgetting SHIFT is one-shot

SHIFT applies only to the next key. If the S indicator is still lit, press SHIFT again to cancel.

Mistake 9: Clearing memory by accident

AC clears the line but not memory; CLR (SHIFT 9) can clear memory and variables.

Mistake 10: Forgetting Reset between problems

Old values (memory, stack, open brackets, worksheet values) can leak into the next calculation. Press Reset or clear before starting a new problem.

Mistake 11: Pressing the wrong layer

A shift / 2nd indicator that is still lit will turn the next key into its shifted function. Look at the display indicators before pressing.

Mistake 12: Minus vs negative

The (−) negative key and the − subtraction key are different. Use (−) to start a negative number, − between two numbers.

9 · Practice drills

Set A — Calculation drills (68)

Work each one on calculator #5, then check your keys and the answer in Appendix A. Every answer was produced by this calculator.

1. 25 + 5 × 6 = _____
2. 7 + 8 × 9 = _____
3. 150 + 12 × 11 = _____
4. 3 + 14 × 15 = _____
5. 48 + 6 × 7 = _____
6. 1000 + 25 × 4 = _____
7. 19 + 21 × 3 = _____
8. 64 + 8 × 8 = _____
9. (45 + 15) ÷ 6 = _____
10. (9 + 7) ÷ 2 = _____
11. (100 + 60) ÷ 8 = _____
12. (33 + 17) ÷ 5 = _____
13. (250 + 350) ÷ 12 = _____
14. (5 + 6) × (10 − 3) = _____
15. (12 + 3) × (8 − 5) = _____
16. (1.5 + 2.5) × (7 − 4) = _____

17. (20 + 30) × (15 − 5) = _____

18. (9 + 1) × (100 − 1) = _____

19. (11 + 13) × (17 − 9) = _____

20. 3^5 = _____

21. 1.05^3 = _____

22. 5^4 = _____

23. 7^3 = _____

24. 1.5^2 = _____

25. 10^6 = _____

26. 2^0.5 = _____

27. −20 + 7 = _____

28. −3 + 3 = _____

29. −100 + 250 = _____

30. −8 + 1 = _____

31. 2×10^3 × 4 = _____

32. 1.5×10^4 × 2 = _____

33. 3×10^5 × 3 = _____

34. 4.5×10^2 × 2 = _____

35. 15 × 6 = , then Ans ÷ 9 = _____

36. 7.5 × 8 = , then Ans ÷ 3 = _____

37. 250 × 12 = , then Ans ÷ 100 = _____

38. (12 + 15 + 18) ÷ 3 = _____

39. (70 + 85 + 91) ÷ 3 = _____

40. (2.5 + 3.5 + 6) ÷ 3 = _____

41. (100 + 200 + 600) ÷ 3 = _____

42. 100 − 3 × 12 + 9 = _____

43. 18 − 2 × 5 + 40 = _____

44. 7.5 − 1.5 × 3 + 2 = _____

45. 64 − 8 × 4 + 10 = _____

46. 300 − 25 × 4 + 55 = _____

47. 9 − 3 × 3 + 1 = _____

48. 45 − 5 × 5 + 15 = _____

49. 1000 ÷ 8 ÷ 5 = _____

50. 96 ÷ 2 ÷ 3 = _____

51. 7200 ÷ 60 ÷ 60 = _____

52. 500 ÷ 4 ÷ 25 = _____

53. 81 ÷ 3 ÷ 3 = _____

54. ((10 + 2) × 3 − 6) ÷ 6 = _____

55. ((1.5 + 2.5) × 6 − 4) ÷ 5 = _____

56. ((20 + 30) × 2 − 25) ÷ 3 = _____

57. ((7 + 8) × 9 − 15) ÷ 12 = _____

58. 12 + 3 × 4 = _____

59. (12 + 8) ÷ 4 = _____

60. (3 + 4) × (9 − 2) = _____

61. 2^10 = _____

62. −5 + 12 = _____

63. 6×10^2 × 3 = _____

64. 100 × 4 = , then Ans ÷ 2 = _____

65. (4 + 8 + 15) ÷ 3 = _____

66. 50 − 4 × 6 + 7 = _____

67. 360 ÷ 4 ÷ 9 = _____

68. ((3 + 5) × 4 − 2) ÷ 10 = _____

Set B — Key-finding drills (24)

Without looking at the table, find each key on the picture. Answers in Appendix A.

1. Which key is “Keyboard” (On-screen keyboard panel)? _____
2. Which key is “Shift” (SHIFT / 2nd)? _____
3. Which key is “ ▲ ” (Cursor ▲)? _____
4. Which key is “ ▼ ” (Cursor ▼)? _____
5. Which key is “◀” (Cursor ◀)? _____
6. Which key is “▶” (Cursor ▶)? _____

7. **Which key is "⌫" (DEL)? _____**
8. **Which key is "Clear" (ON / AC)? _____**
9. **Which key is "=" (Equals)? _____**
10. **Which key is "x" (Variable X)? _____**
11. **Which key is "y" (Variable Y)? _____**
12. **Which key is "z" (Variable Z)? _____**
13. **Which key is "^" (Power x^y / ^)? _____**
14. **Which key is "÷" (Division ÷)? _____**
15. **Which key is "(" (Opening bracket)? _____**
16. **Which key is "×" (Multiplication ×)? _____**
17. **Which key is ")" (Closing bracket). Closing brackets at the end of a line may be omitted)? _____**
18. **Which key is "−" (Subtraction −)? _____**
19. **Which key is "," (Comma / argument separator)? _____**
20. **Which key is "+" (Addition +)? _____**
21. **Which key is "(−)" (Negative sign)? _____**
22. **Which key is "." (Decimal point. Pressing it first gives "0."; a second point in the same number is ignored)? _____**
23. **Which key is "EXP" (×10^x / EXP / EE)? _____**
24. **Which key is "EXE" (Execute / =)? _____**

Appendix A — Drill answers

Set A

1. 55 (display `55`)

`#27` 2 → `#22` 5 → `#29` + → `#22` 5 → `#19` × → `#23` 6 → `#34` EXE

2. 79 (display `79`)

`#16` 7 → `#29` + → `#17` 8 → `#19` × → `#18` 9 → `#34` EXE

3. 282 (display `282`)

`#26` 1 → `#22` 5 → `#31` 0 → `#29` + → `#26` 1 → `#27` 2 → `#19` × → `#26` 1 → `#26` 1 → `#34` EXE

4. 213 (display `213`)

`#28` 3 → `#29` + → `#26` 1 → `#21` 4 → `#19` × → `#26` 1 → `#22` 5 → `#34` EXE

5. 90 (display `90`)

`#21` 4 → `#17` 8 → `#29` + → `#23` 6 → `#19` × → `#16` 7 → `#34` EXE

6. 1100 (display `1100`)

`#26` 1 → `#31` 0 → `#31` 0 → `#31` 0 → `#29` + → `#27` 2 → `#22` 5 → `#19` × → `#21` 4 → `#34` EXE

7. 82 (display 82)

#26 1 → #18 9 → #29 + → #27 2 → #26 1 → #19 × → #28 3 → #34 EXE

8. 128 (display 128)

#23 6 → #21 4 → #29 + → #17 8 → #19 × → #17 8 → #34 EXE

9. 10 (display 10)

#15 (→ #21 4 → #22 5 → #29 + → #26 1 → #22 5 → #20) → #14 ÷ → #23 6 → #34 EXE

10. 8 (display 8)

#15 (→ #18 9 → #29 + → #16 7 → #20) → #14 ÷ → #27 2 → #34 EXE

11. 20 (display 20)

#15 (→ #26 1 → #31 0 → #31 0 → #29 + → #23 6 → #31 0 → #20) → #14 ÷ → #17 8 → #34 EXE

12. 10 (display 10)

#15 (→ #28 3 → #28 3 → #29 + → #26 1 → #16 7 → #20) → #14 ÷ → #22 5 → #34 EXE

13. 50 (display `50`)

#15 (→ #27 2 → #22 5 → #31 0 → #29 + → #28 3 → #22 5 → #31 0 → #20) → #14 ÷ → #26 1 → #27 2 → #34 EXE

14. 77 (display `77`)

#15 (→ #22 5 → #29 + → #23 6 → #20) → #19 × → #15 (→ #26 1 → #31 0 → #24 − → #28 3 → #20) → #34 EXE

15. 45 (display `45`)

#15 (→ #26 1 → #27 2 → #29 + → #28 3 → #20) → #19 × → #15 (→ #17 8 → #24 − → #22 5 → #20) → #34 EXE

16. 12 (display `12`)

#15 (→ #26 1 → #32 . → #22 5 → #29 + → #27 2 → #32 . → #22 5 → #20) → #19 × → #15 (→ #16 7 → #24 − → #21 4 → #20) → #34 EXE

17. 500 (display `500`)

#15 (→ #27 2 → #31 0 → #29 + → #28 3 → #31 0 → #20) → #19 × → #15 (→ #26 1 → #22 5 → #24 − → #22 5 → #20) → #34 EXE

18. 990 (display `990`)

`#15` (→ `#18` 9 → `#29` + → `#26` 1 → `#20`) → `#19` × → `#15` (→ `#26` 1 → `#31` 0 → `#31` 0 → `#24` − → `#26` 1 → `#20`) → `#34` EXE

19. 192 (display `192`)

`#15` (→ `#26` 1 → `#26` 1 → `#29` + → `#26` 1 → `#28` 3 → `#20`) → `#19` × → `#15` (→ `#26` 1 → `#16` 7 → `#24` − → `#18` 9 → `#20`) → `#34` EXE

20. 243 (display `243`)

`#28` 3 → `#13` ^ → `#22` 5 → `#34` EXE

21. 1.157625 (display `1.157625`)

`#26` 1 → `#32` . → `#31` 0 → `#22` 5 → `#13` ^ → `#28` 3 → `#34` EXE

22. 625 (display `625`)

`#22` 5 → `#13` ^ → `#21` 4 → `#34` EXE

23. 343 (display `343`)

`#16` 7 → `#13` ^ → `#28` 3 → `#34` EXE

24. 2.25 (display `2.25`)

`#26` 1 → `#32` . → `#22` 5 → `#13` ^ → `#27` 2 → `#34` EXE

25. 1000000.000 (display `1000000`)

`#26` 1 → `#31` 0 → `#13` ^ → `#23` 6 → `#34` EXE

26. 1.41421356 (display `1.414213562`)

`#27` 2 → `#13` ^ → `#31` 0 → `#32` . → `#22` 5 → `#34` EXE

27. -13 (display `-13`)

`#30` (−) → `#27` 2 → `#31` 0 → `#29` + → `#16` 7 → `#34` EXE

28. 0 (display `0`)

`#30` (−) → `#28` 3 → `#29` + → `#28` 3 → `#34` EXE

29. 150 (display `150`)

`#30` (−) → `#26` 1 → `#31` 0 → `#31` 0 → `#29` + → `#27` 2 → `#22` 5 → `#31` 0 → `#34` EXE

30. -7 (display `-7`)

`#30` (−) → `#17` 8 → `#29` + → `#26` 1 → `#34` EXE

31. 8000 (display `8000`)

`#27` 2 → `#33` EXP → `#28` 3 → `#19` × → `#21` 4 → `#34` EXE

32. 30000 (display `30000`)

`#26` 1 → `#32` . → `#22` 5 → `#33` EXP → `#21` 4 → `#19` × → `#27` 2 → `#34` EXE

33. 900000 (display `900000`)

`#28` 3 → `#33` EXP → `#22` 5 → `#19` × → `#28` 3 → `#34` EXE

34. 900 (display `900`)

`#21` 4 → `#32` . → `#22` 5 → `#33` EXP → `#27` 2 → `#19` × → `#27` 2 → `#34` EXE

35. 10 (display `10`)



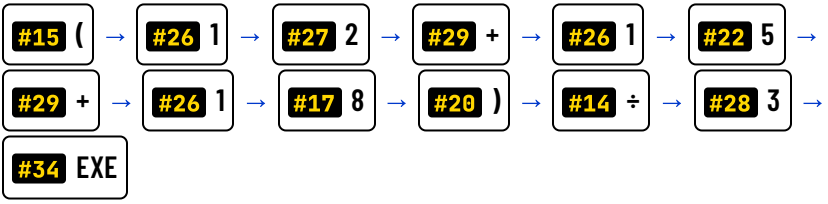
36. 20 (display `20`)



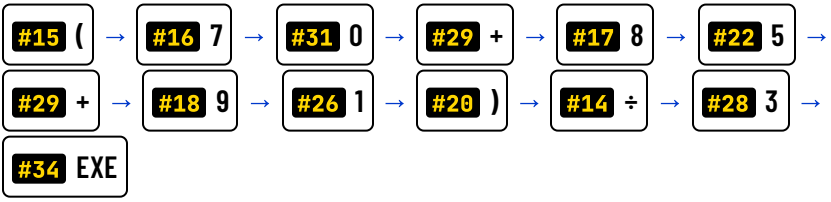
37. 30 (display `30`)



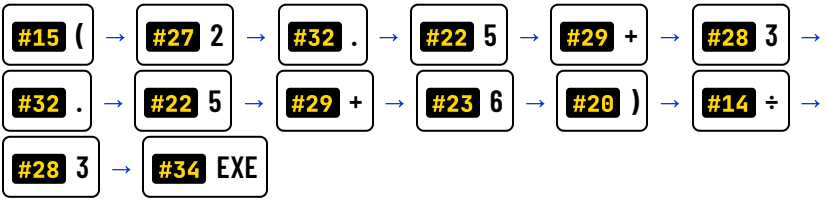
38. 15 (display `15`)



39. 82 (display `82`)



40. 4 (display `4`)



41. 300 (display 300)

#15 (→ #26 1 → #31 0 → #31 0 → #29 + → #27 2 →
#31 0 → #31 0 → #29 + → #23 6 → #31 0 → #31 0 →
#20) → #14 ÷ → #28 3 → #34 EXE

42. 73 (display 73)

#26 1 → #31 0 → #31 0 → #24 − → #28 3 → #19 × →
#26 1 → #27 2 → #29 + → #18 9 → #34 EXE

43. 48 (display 48)

#26 1 → #17 8 → #24 − → #27 2 → #19 × → #22 5 →
#29 + → #21 4 → #31 0 → #34 EXE

44. 5 (display 5)

#16 7 → #32 . → #22 5 → #24 − → #26 1 → #32 . →
#22 5 → #19 × → #28 3 → #29 + → #27 2 → #34 EXE

45. 42 (display 42)

#23 6 → #21 4 → #24 − → #17 8 → #19 × → #21 4 →
#29 + → #26 1 → #31 0 → #34 EXE

46. 255 (display 255)

#28 3 → #31 0 → #31 0 → #24 − → #27 2 → #22 5 →
#19 × → #21 4 → #29 + → #22 5 → #22 5 → #34 EXE

47. 1 (display 1)

#18 9 → #24 − → #28 3 → #19 × → #28 3 → #29 + → #26 1 → #34 EXE

48. 35 (display 35)

#21 4 → #22 5 → #24 − → #22 5 → #19 × → #22 5 → #29 + → #26 1 → #22 5 → #34 EXE

49. 25 (display 25)

#26 1 → #31 0 → #31 0 → #31 0 → #14 ÷ → #17 8 → #14 ÷ → #22 5 → #34 EXE

50. 16 (display 16)

#18 9 → #23 6 → #14 ÷ → #27 2 → #14 ÷ → #28 3 → #34 EXE

51. 2 (display 2)

#16 7 → #27 2 → #31 0 → #31 0 → #14 ÷ → #23 6 → #31 0 → #14 ÷ → #23 6 → #31 0 → #34 EXE

52. 5 (display 5)

#22 5 → #31 0 → #31 0 → #14 ÷ → #21 4 → #14 ÷ → #27 2 → #22 5 → #34 EXE

53. 9 (display 9)

#17 8 → #26 1 → #14 ÷ → #28 3 → #14 ÷ → #28 3 → #34 EXE

54. 5 (display 5)

#15 (→ #15 (→ #26 1 → #31 0 → #29 + → #27 2 → #20) → #19 × → #28 3 → #24 − → #23 6 → #20) → #14 ÷ → #23 6 → #34 EXE

55. 4 (display 4)

#15 (→ #15 (→ #26 1 → #32 . → #22 5 → #29 + → #27 2 → #32 . → #22 5 → #20) → #19 × → #23 6 → #24 − → #21 4 → #20) → #14 ÷ → #22 5 → #34 EXE

56. 25 (display 25)

#15 (→ #15 (→ #27 2 → #31 0 → #29 + → #28 3 → #31 0 → #20) → #19 × → #27 2 → #24 − → #27 2 → #22 5 → #20) → #14 ÷ → #28 3 → #34 EXE

57. 10 (display 10)

#15 (→ #15 (→ #16 7 → #29 + → #17 8 → #20) → #19 × → #18 9 → #24 − → #26 1 → #22 5 → #20) → #14 ÷ → #26 1 → #27 2 → #34 EXE

58. 24 (display 24)

#26 1 → #27 2 → #29 + → #28 3 → #19 × → #21 4 → #34 EXE

59. 5 (display `5`)

`#15` (→ `#26` 1 → `#27` 2 → `#29` + → `#17` 8 → `#20`) → `#14` ÷ → `#21` 4 → `#34` EXE

60. 49 (display `49`)

`#15` (→ `#28` 3 → `#29` + → `#21` 4 → `#20`) → `#19` × → `#15` (→ `#18` 9 → `#24` − → `#27` 2 → `#20`) → `#34` EXE

61. 1024 (display `1024`)

`#27` 2 → `#13` ^ → `#26` 1 → `#31` 0 → `#34` EXE

62. 7 (display `7`)

`#30` (−) → `#22` 5 → `#29` + → `#26` 1 → `#27` 2 → `#34` EXE

63. 1800 (display `1800`)

`#23` 6 → `#33` EXP → `#27` 2 → `#19` × → `#28` 3 → `#34` EXE

64. 200 (display `200`)

`#26` 1 → `#31` 0 → `#31` 0 → `#19` × → `#21` 4 → `#34` EXE → `#14` ÷ → `#27` 2 → `#34` EXE

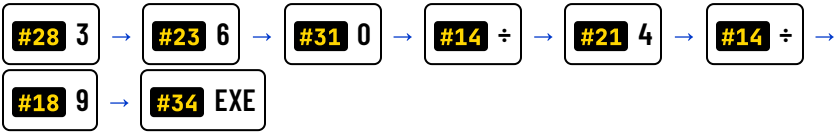
65. 9 (display `9`)

`#15` (→ `#21` 4 → `#29` + → `#17` 8 → `#29` + → `#26` 1 → `#22` 5 → `#20`) → `#14` ÷ → `#28` 3 → `#34` EXE

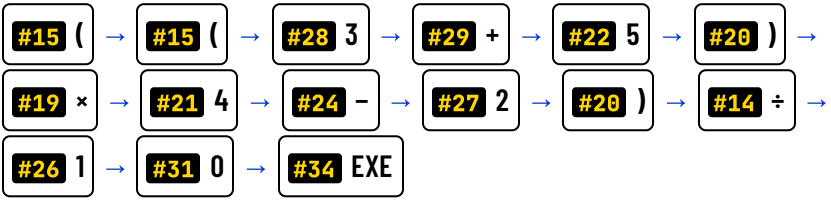
66. 33 (display `33`)

`#22` 5 → `#31` 0 → `#24` − → `#21` 4 → `#19` × → `#23` 6 → `#29` + → `#16` 7 → `#34` EXE

67. 10 (display `10`)



68. 3 (display `3`)



Set B

1. #1
2. #2
3. #3
4. #4
5. #5
6. #6
7. #7
8. #8
9. #9
10. #10
11. #11
12. #12
13. #13
14. #14
15. #15
16. #19
17. #20
18. #24
19. #25
20. #29
21. #30
22. #32
23. #33
24. #34

Appendix B — Function finder (A–Z)

Function	Label	Keys
×10ˣ / EXP / EE — enter a power-of-ten exponent (scientific notation), e.g. 6.02 ×10ˣ 23	EXP	#33
Addition +	+	#29
Closing bracket). Closing brackets at the end of a line may be omitted	)	#20
Comma / argument separator (between function arguments, list items or coordinates)	,	#25
Cursor ▲ — up / previous history line / previous item	▲	#3
Cursor ▶ — move right in the line	▶	#6
Cursor ▼ — down / next history line / next item	▼	#4
Cursor ◀ — move left in the line	◀	#5
Decimal point. Pressing it first gives "0."; a second point in the same number is ignored	.	#32
DEL — delete the character at the cursor	⌫	#7
Digit 0	0	#31
Digit 1	1	#26
Digit 2	2	#27
Digit 3	3	#28
Digit 4	4	#21
Digit 5	5	#22
Digit 6	6	#23
Digit 7	7	#16
Digit 8	8	#17
Digit 9	9	#18
Division ÷	÷	#14
Equals — complete the calculation	=	#9
Execute / = — evaluate the line and show the result (EXE / ENTER / =)	EXE	#34

Multiplication ×	×	#19
Negative sign (−) — makes the next number negative. Different from the subtraction key	(−)	#30
ON / AC — power on, or clear all	Clear	#8
On-screen keyboard panel	Keyboard	#1
Opening bracket (	(	#15
Power x^y / ^ — raise to any power. Chains right-to-left (2^3^2 = 2^9 = 512)	^	#13
SHIFT / 2nd — the next key uses its upper (shift-coloured) function	Shift	#2
Subtraction −	−	#24
Variable X (store with STO/→, recall by name)	x	#10
Variable Y (store with STO/→, recall by name)	y	#11
Variable Z (store with STO/→, recall by name)	z	#12

Appendix C — Glossary

Ans
The result of the last calculation, reusable in the next.

AOS
Algebraic Operating System — operator precedence on a one-line display.

RPN
Reverse Polish Notation — operands first, then the operation; no = key.

Stack
The RPN registers X, Y, Z, T where numbers wait for operations.

TVM
Time Value of Money — n, i, PV, PMT, FV.

Natural display
Shows fractions, roots and powers as printed in textbooks.

Shift / 2nd
Prefix key that selects the function printed above a key.

Alpha
Prefix key that types the letter printed beside a key.

Radian
Angle unit; 2π rad = 360°.

Grad
Angle unit; 400 grad = 360°.

Engineering notation
Scientific notation with exponents that are multiples of 3.

Mantissa
The digits of a number in scientific notation.

Memory (M)

An independent register for running totals.

Mark-up

Increase over cost; margin is profit as a percent of selling price.

Grand total (GT)

Sum of all totals since the last clear.

nCr / nPr

Combinations / permutations of n items taken r at a time.

Factorial

n! = 1×2×…×n.

Reciprocal

1/x.

Standard deviation

Spread of data around the mean (σ population, s sample).

Key number

The small badge 1…N on each key in U-45 CALCULATORS, used throughout this manual.

Appendix D — Model notes & limits

Touch-screen colour model. Tap "Keyboard" to toggle the on-screen soft keyboard (functions, templates, variables). The touch screen menu bar items and app icons are tappable; the stylus-driven geometry, eActivity, spreadsheet and CAS algebra (solve/factor/expand symbolic) are simplified to numeric work.

Planned (not yet working)

- Symbolic CAS algebra (solve, factor, expand)
- Geometry app
- eActivity
- Spreadsheet
- Stylus drawing
- Graph image export (PNG)
- Shareable function list link
- Dark screen theme
- Speech read-out of values
- Exam mode

U-45 CALCULATORS models are independent software recreations; they are not affiliated with any calculator manufacturer and carry no third-party trademarks.

Appendix E — Quick-reference card



#1 Keyboard	#2 Shift	#3 ▲
#4 ▼	#5 ◀	#6 ▶
#7 ⌫	#8 Clear	#9 =
#10 x	#11 y	#12 z
#13 ^	#14 ÷	#15 (
#16 7	#17 8	#18 9
#19 ×	#20)	#21 4
#22 5	#23 6	#24 −
#25 ,	#26 1	#27 2
#28 3	#29 +	#30 (−)
#31 0	#32 .	#33 EXP
#34 EXE		

U-45 CALCULATORS · Calculator #5 · cut out and keep